

Sales Forecasting model...

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt


from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score


# -----

# 1. Create Sample Sales Dataset

# -----

np.random.seed(42)


dates = pd.date_range(start="2024-01-01", periods=200,
freq="D")


marketing_spend = np.random.uniform(100, 1000, 200)
```

```
footfall = marketing_spend * 1.5 + np.random.normal(50,  
30, 200)
```

```
sales = (  
    marketing_spend * 2.5  
    + footfall * 1.2  
    + np.random.normal(200, 100, 200)  
)
```

```
data = pd.DataFrame({  
    "Date": dates,  
    "Marketing_Spend": marketing_spend,  
    "Footfall": footfall,  
    "Sales": sales  
})
```

```
# Add missing values
```

```
data.loc[15:20, "Marketing_Spend"] = np.nan
```

```
print("Original Data:")
```

```
print(data.head())
```

```
# -----
```

```
# 2. Data Cleaning
```

```
# -----
```

```
data["Marketing_Spend"] =  
data["Marketing_Spend"].fillna(  
    data["Marketing_Spend"].mean()  
)
```

```
# Feature Engineering
```

```
data["Day_of_Week"] = data["Date"].dt.dayofweek  
data["Month"] = data["Date"].dt.month
```

```
# Features and Target
```

```
X = data[["Marketing_Spend", "Footfall", "Day_of_Week",  
"Month"]]  
y = data["Sales"]
```

```
# Train-Test Split
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42
```

)

```
print("\nTraining Samples:", len(X_train))
```

```
print("Testing Samples:", len(X_test))
```

```
# -----
```

```
# 3. Train Model
```

```
# -----
```

```
model = RandomForestRegressor(
```

```
    n_estimators=100,
```

```
    random_state=42
```

```
)
```

```
model.fit(X_train, y_train)
```

```
print("\nModel Trained Successfully!")
```

```
# -----
```

```
# 4. Prediction
```

```
# -----
```

```
y_pred = model.predict(X_test)
```

```
# Evaluation
```

```
mae = mean_absolute_error(y_test, y_pred)
```

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("\nModel Evaluation")
```

```
print("MAE :", round(mae, 2))
```

```
print("RMSE:", round(rmse, 2))
```

```
print("R2 Score:", round(r2, 4))
```

```
# -----
```

```
# 5. Visualization
```

```
# -----
```

```
plt.figure(figsize=(12, 6))
```

```
plt.plot(
```

```
    y_test.values,
```

```
    label="Actual Sales",
```

```
    marker="o"
```

```
)
```

```
plt.plot(  
    y_pred,  
    label="Predicted Sales",  
    linestyle="--",  
    marker="x"  
)  
  
plt.title("Actual Sales vs Predicted Sales")  
plt.xlabel("Test Data Points")  
plt.ylabel("Sales Revenue")  
plt.legend()  
plt.grid(True)  
  
plt.tight_layout()  
plt.savefig("sales_forecast_trend.png")  
plt.show()
```

RESULT...

Console



Run

Original Data:

	Date	Marketing_Spend	Footfall	Sales
0	2024-01-01	437.086107	685.228419	2200.755332
1	2024-01-02	955.642876	1490.431925	4361.631646
2	2024-01-03	758.794548	1196.983996	3531.465543
3	2024-01-04	638.792636	986.758411	2880.838746
4	2024-01-05	240.416776	466.598400	1359.108707

Training Samples:

160

Testing Samples: 40

Model Trained Successfully!

Model Evaluation

MAE : 119.8

RMSE: 146.67

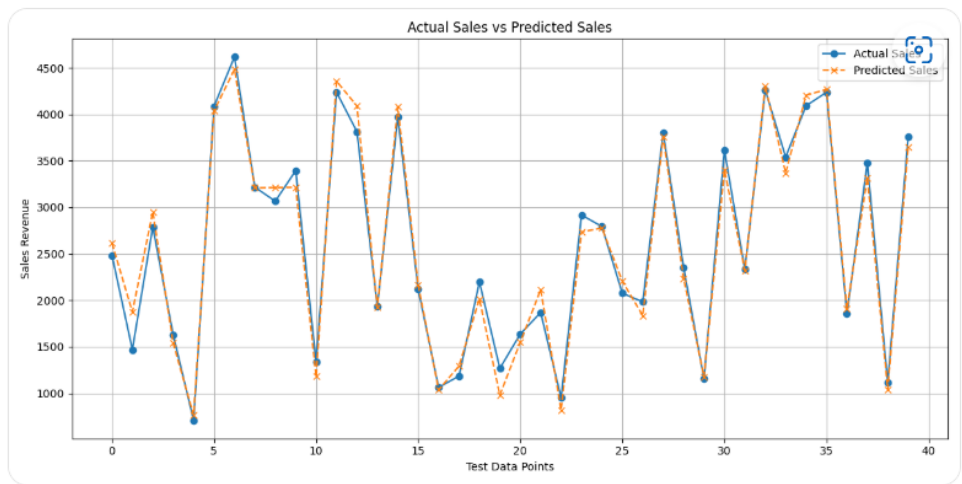
R² Score: 0.9831

Model Evaluation

MAE : 119.8

RMSE: 146.67

R² Score: 0.9831



Run completed in 73766ms

